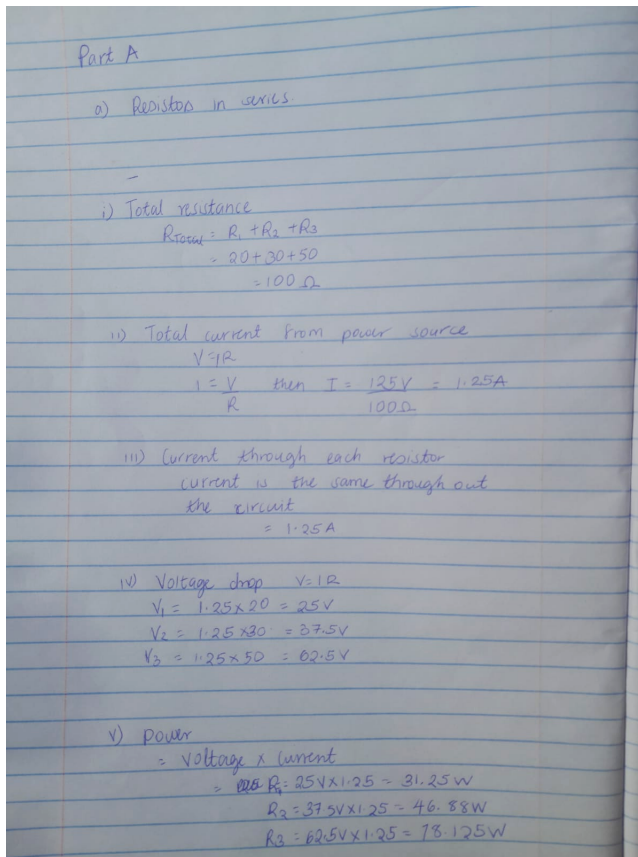


Lab: Schematics with ESP32
153356

Part A

Calculations



1. Calculate the total resistance in ohms,

$R_{\text{total}} = R_1 + R_2 + R_3$

100 ohms

2. Calculate the total current from the power supply,

1.25A

3. Calculate the current through each resistor

1.25 - current is the same throughout the circuit

4. Calculate the Voltage drop through each resistor

R1 = 25V

R2 = 37.5V

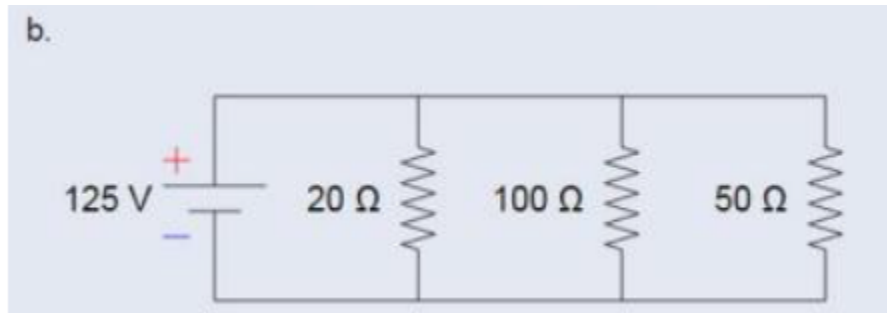
R3 = 62.5V

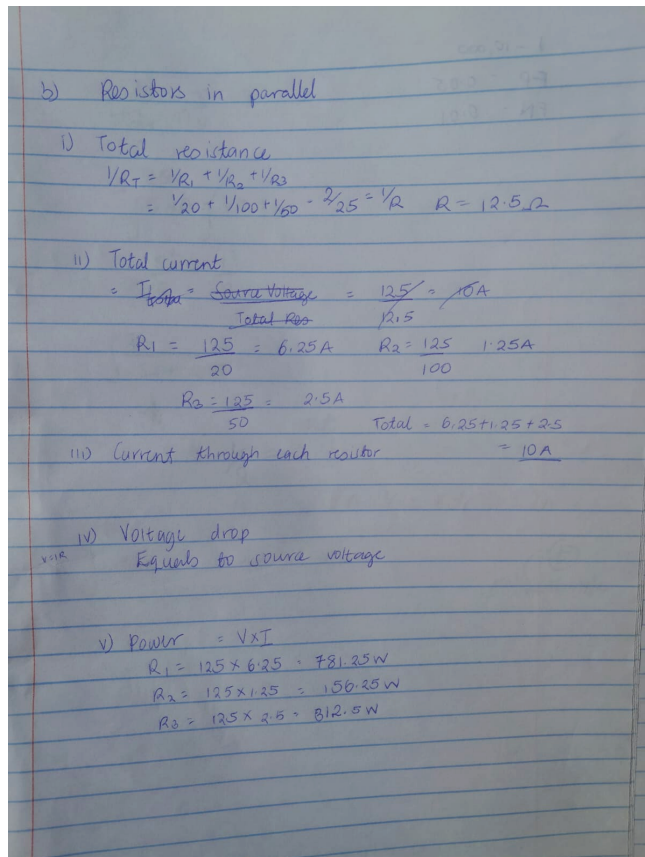
5. Calculate the Power dissipated through each resistor (use Watt's Law: Power = Voltage * Current)

R1 = 31.25 W

R2 = 46.88W

R3 = 78.125W





1. Calculate the total resistance in ohms,

$$\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$= 12.5 \text{ ohms}$$

2. Calculate the total current from the power supply,

$$\text{Total voltage} / \text{total current} = 125V / 12.5 = 10A$$

3. Calculate the current through each resistor

$$R_1 = 125 / 20 = 6.25A$$

$$R_2 = 125 / 100 = 1.25A$$

$$R_3 = 125 / 50 = 2.5A$$

4. Calculate the Voltage drop through each resistor

$$\text{Voltage is equal to the source through the circuit} = 125V$$

5. Calculate the Power dissipated through each resistor (use Watt's Law: Power = Voltage * Current

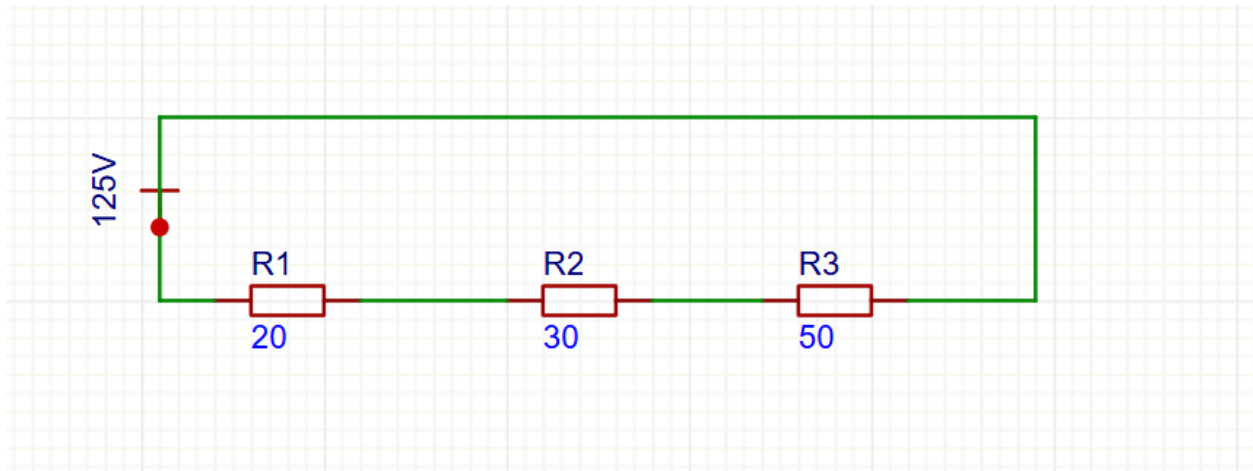
$$R_1 = 125 \times 6.25 = 781.25W$$

$$R_2 = 125 \times 1.25 = 156.25W$$

$$R_3 = 125 \times 2.5 = 312.5W$$

Part B

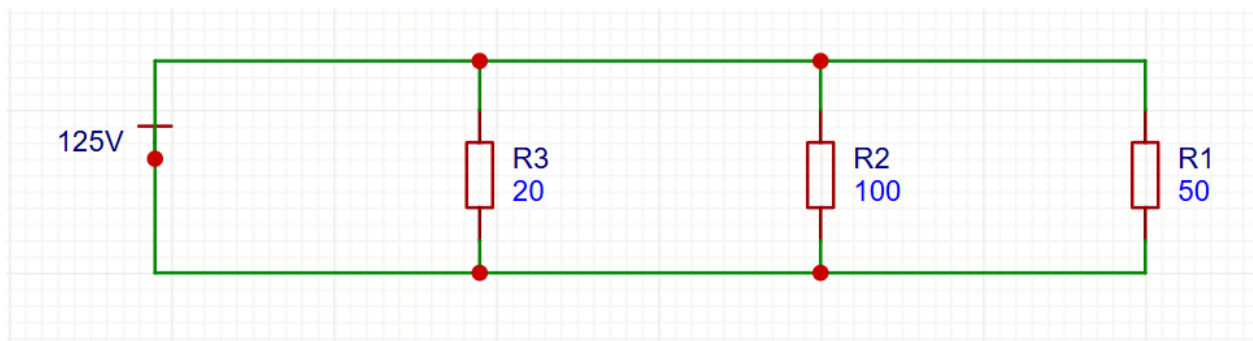
Schematic a



List of components

	A	B	C	D	E	F	G	H	I	J
1	No.	Quantity	Comment	Designator	Footprint	Value	Manufacturer Part	Manufacturer	Supplier Part	Supplier
2	1	1	20	R1	R0603	20				
3	2	1	30	R2	R0603	30				
4	3	1	50	R3	R0603	50				
5										

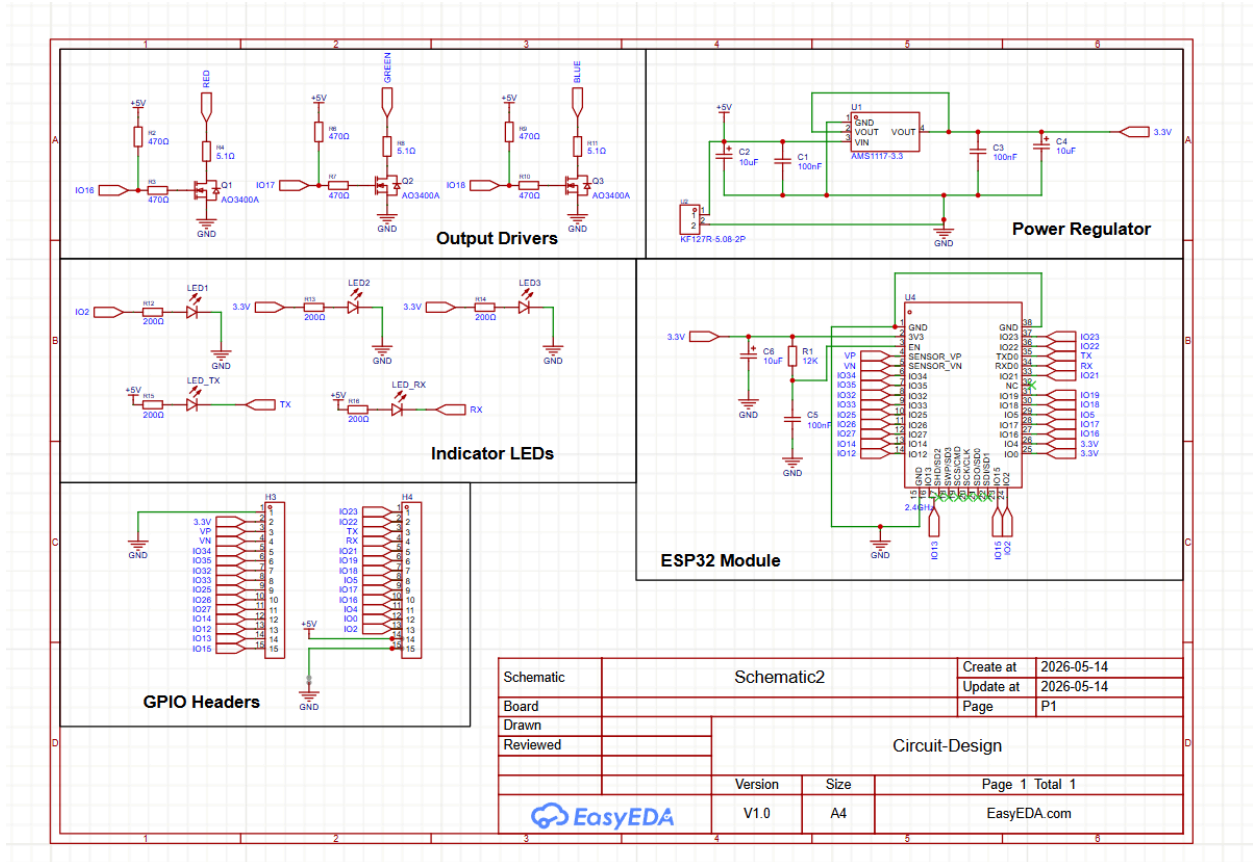
Schematic b



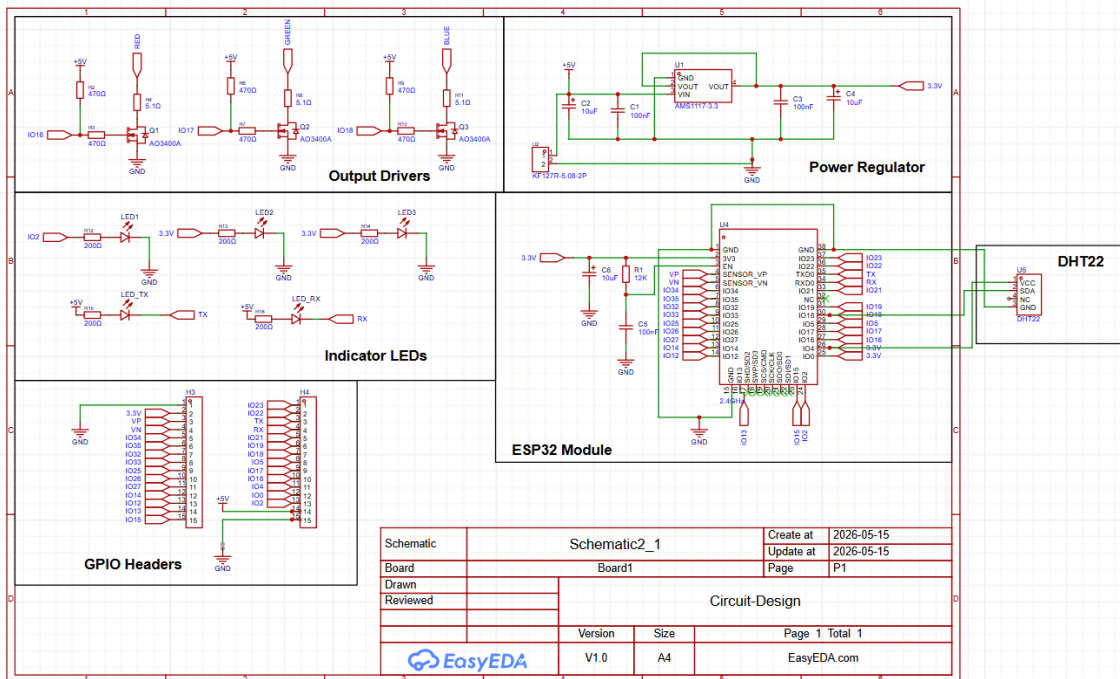
List of components

	A	B	C	D	E	F	G	H	I	J
1	No.	Quantity	Comment	Designator	Footprint	Value	Manufacturer Part	Manufacturer	Supplier Part	Supplier
2	1	1	50	R1	R0603	50				
3	2	1	100	R2	R0603	100				
4	3	1	20	R3	R0603	20				
5										

Schematic diagram for ESP32



With DHT22 connection



List of components

[illegible]